Research Assignment Part 2

1. Business Analytics - is the process of using statistical methods, computing technologies, and data mining to transform raw data into actionable insights for improved decision-making.
- Retail inventory management - a clothing chain analyses sales data, ~~wha~~ weather patterns, and local events to forecast inventory needs

2. Data Analytics - is the process of collecting, cleaning, and analysing raw data to uncover trends, patterns, and insights that inform better decision-making.
- Customer Segmentation - E-commerce platforms analyse browsing and purchasing habits to target specific customer groups with personalised offers

3. Business intelligence - using software, tools, and methodologies to transform raw data into actionable insights, helping organisations track performance and make informed, data driven decisions.
- Finance management - banks use BI to analyse client financial data in real time, enabling them to evaluate risk during loan approvals.

4. Data-Driven Decision Making - is the process of using insights derived from data analysis, rather than intuition, to guide business strategies and actions.
- Healthcare - hospitals analyse patient data to identify high risk individuals and reduce readmission rates by providing early intervention

5. Big data - is the process of examining massive, complex datasets to uncover hidden patterns, correlations, and insights that inform better business decisions

- fraud detection - banks such as American Express analyse transaction patterns in real-time to detect suspecious activity.

6. Descriptive analytics - is the foundational process of analysing historical and current data to understand what happened in a business, often using dashboards, reports, and visualisation tools

- Customer behaviour - summarizing monthly revenue, expenses, and cash flow to assess business health

7. Diagnostic analytics - a form of data analysis that examines data to understand the root causes of trends and anomalies, answering the question "What did this happen?"

- Customer Churn Analysis - investigating why customers canceled subscriptions.

8. Predictive analytics - a branch of advanced data science that uses historical data, statistical modeling, and machine learning techniques to forecast future outcomes

- Custome Churn Prevention - identifying users likely to cancel subscriptions, allowing for targeted retention offers

9. Prescriptive analytics - is the final, most advanced stage of data analysis that recommends specific actions

to achieve a desired outcome, answering "What should we do about it?"

- Dynamic pricing - airlines and retailers automatically adjust prices based on demand, competition, and inventory to maximise revenue.

10. Data literacy - is the ability to read, work with, analyse, and communicate with data to make informed decisions.

- Data driven sales - a sales representative uses CRM data to identify which leads are most likely to convert, rather than guessing

11. Structured data - is highly organised information, typically in tables with rows and columns, making it easily searchable and analysed by machines.

- Financial transaction - a bank database tracking account numbers, transaction dates, and amounts, enabling instant daily balance reporting.

12. Unstructured data - is information that lacks a predefined, rigid model, making it difficult to store in traditional rows and columns.

- Customer Sentiment Analysis - retailers use Natural Language Processing to analyse social media comments, reviews, and support chat logs to determine customer satisfaction.

13. Data warehouse - is a centralised repository that consolidates structured data from multiple sources like CRMs, ERPs, and SQL databases

into one consistent platform for reporting and analysis.

- Retail - a retailer combines data from online sales, in store transactions, and loyalty programs to analyse regional demands trends, optimise inventory levels, and identify high-value customer segments.

14. Data mining is the process of discovering hidden patterns, anomalies, and correlations with large datasets using machine learning, statistics, and database system to turn raw data into actionable insights.

- Textie mining - analysing customer reviews, social media, and emails to understand customer sentiment

15. Data visualisation - is the graphical representation of information transforming raw data into charts, graphs, or maps to identify patterns and insights.

- Business dashboards - Tableau or Power BI dashbeards tracking real-time like sales revenue, customer churn, and marketing performance.

16. Dashboard - is a visual tool that centralises data, using charts, graphs, and KPIs to monitor performance and support decision-making.

- Sales dashboard - visualise quarterly sales targets, revenue breakdown by product, and customer engagement metrics.

17. Metrics - are quantifiable, raw data points tracking performance, while analytics is the process of interpreting that data to understand trends and drive decisions.

- Website - ~~M~~ metric (bounce rate, pageviews), analytics (determining that high bounce rates are due to poor mobile optimisation)

18. Key Performance Indicator - is a measurable value that demonstrates how effectively a company is achieving key business objectives.

- Financial: Net Profit Margin (Percentage of revenue that is profit) or ROI (Return on Investment)

19. Benchmarking in analytics is the process of measuring a company's performance metrics against industry standards, competitors, or internal historical data

- Performance tracking - a software company analyse its app loading time against competitors to improve user experience.

20. Segmentation is the process of dividing a broad dataset, typically customers or website users into smaller, managable subgroups based on shared characteristics like behaviour, demographics, or location.

- Behavioral segmentation - is grouping users by actions, such as targeting users who added items to a ~~chart~~ cart but didn't purchase with a discount email.

21. Conversion rate - is the percentage of users who complete a specific, desired action, such as buying a product, signing up for a newsletter, or filling out a form out of the total number of visitors or participants

- E-Commerce - If 100 000 people visit an online store in April and 2000 purchase a product, the conversion rate is $\left(\frac{2000}{100\,000} \times 100\right) = 2\%$

22 Customer Acquisition Cost (CAC) - is a key analytics metric measuring the total average expenses, including marketing, advertising, and sales staff salaries, required to acquire a single new customer

- Performance Tracking - comparing the efficiency of different channels to allocate budget to the lowest CAC channels

23 Customer Lifetime Value (CLV/LTV) - measures the total revenue or net profit a business can expect from a single customer account throughout their entire relationship

- Marketing ROI - comparing LTV to Customer Acquisition Cost (CAC) to determine profitability, typically looking for ratio 3:1

24 Churn rate - is the percentage of customers, suberibers or users who stop doing business with an entity during a specific time frame

- Example a software company, starts the month of January with 1 000 customers. During

January, 50 customers cancel their subscriptions

Customer churn $= \frac{50}{1000} \times 100$

$= 5\%$

∴ The company has a 5% monthly churn rate

25. Customer Retention Rate - in analytics measures the percentage of existing customers a company retains over a specific period, reflecting loyalty and business stability.

Example:

* Start of $Q_1 = 100$ customers
* New customers = 20
* End of $Q_1 = 95$ customers

∴ Original number of customers $= 95 - 20$

$= 75$

∴ Customer Retention Rate $= \frac{95-20}{100} \times 100$

$= 75\%$

26. Net Promoter Score - is a customer loyalty metric measuring a user's willingness to recommend a company's product or service.

- Survey question - on a scale of 0-10, how likely are you to recommend us to a friend.

27. Average Order Value - measures the average amount a customer spends per transaction, calculated by dividing total revenue by the number of orders

- Segmenting Traffic Sources - analysing if customers from

paid social campaigns have a higher AOV than those from email marketing to determine channel profitability.

~~28. Sales f~~

28. Sales funnel - is a visual, step by step model mapping the customer journey from initial awareness to final purchase.

- E-commerce - tracking the flow from product page view, add chart, initiate checkout, purchase

29. Marketing campaign - is the systematic tracking, measurement, and analysis of data from marketing initiatives across various channels to evaluate performance, optimise ROI and understand audience behaviour.

- A/B Testing - comparing two versions of an email or landing page to see which performs better regarding click-through rates (CTR)

30. Post-Campaign Analysis (PCA) - is the evaluation of marketing data after a campaign concludes to determine effectiveness calculate ROI, and gain insights for future efforts.

- Example: A company runs a 4-week facebook ad campaign for a new product

Analyse the "session campaign" report in GA4 to identify which ad creative drove the most sale.

31. Incremental revenue - is the additional income generated directly from a specific action such as marketing campaign calculated as the revenue earned

during the action minus the baseline revenue that would have occurred without it.

- Pricing strategy - determining the impact on revenue when raising or lowering prices, allowing for the evaluation of the new price point's efficiency.

32. Return on Investment - measures profitability or efficiency of data-driven projects such as marketing campaigns

- Example:

Marketing campaign - a company spends R5 000 on a digital ad and that campaign generates R8 000 in new revenue

$$\text{Calculation} = \frac{R8\,000 - R5\,000}{R5\,000} \times 100$$

$$ROI = 60\%$$

33. Return on Ad Spend - is a marketing metric measuring revenue generated for every amount spent on advertising, calculated as Revenue ÷ Cost

34. Click Through Rate - measures the percentage of people who click on a link after seeing it, calculated as (Clicks ÷ Impressions) × 100.

- Example

Search engine ads - if an ad shows 1 000 times (impressions) and gets 30 clicks, the CTR is 3%. The higher CTR here often lowers costs and boosts ad position.

35. Bounce rate - is the percentage of single-page sessions in which a visitor leaves your site from the landing page without interacting further.

- Example

If a landing page receives 1000 visitors, and 800 of them leave after viewing only that page

- $\frac{500}{1000} \times 100$

Bounce rate = 50%

36. AB Testing is a controlled experiment where two versions of a ~~dit~~ digital asset are shown to random user segments to compare performance such as conversion rate

- Example: Website CTA Optimisation, company tests a blue "Buy Now" button A against a green "Order Today" button (B) to see which drives more clicks

37. Lead generation - is the process of using data-driven insights, tracking tools, and marketing strategies to identify, attract, and convert potential customers who have shown interest in a product or service.

- Free trails - The sign-up process captures contact details, turning a visitor into a Product Qualified Lead

38. Marketing attribution is the process of identifying which marketing touchpoints such as ad clicks, emails, or social media posts drive conversions

and assigning value to them.

-first clicks - best for measuring awareness